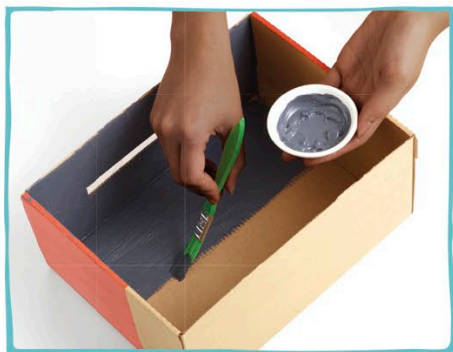




3 Paint the box gray to make it look cavelike. If you want to be really geological, paint streaks in other colors to represent minerals in the rocks.



4 Mix the food coloring with the warm water, then pour it into both glasses. Add Epsom salt to both glasses and stir until no more salt will dissolve.



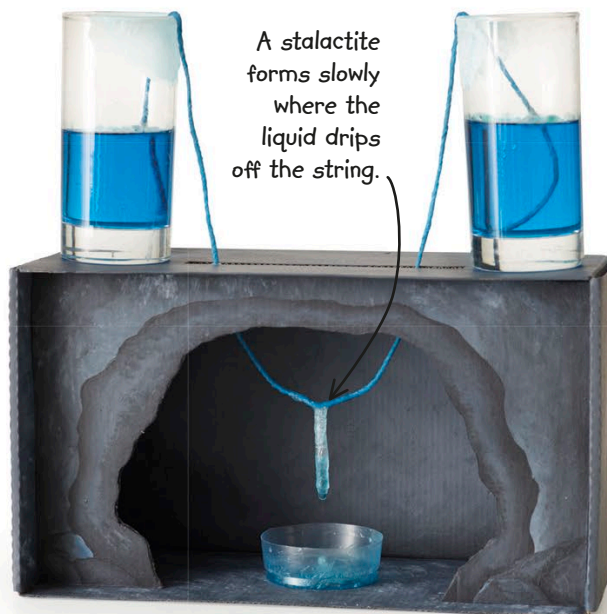
5 Cut off the bottom of the plastic cup to make a shallow bowl. This will catch drips of liquid, as the setup in the next step shows you.



Wet the whole length of the string before you hang it from the glasses.

6 Cut 16in (40cm) of string and put about 4in (10cm) of each end in each glass. Place a glass on each end of the box top, with the middle of the string hanging in a "V" shape through the slit.

7 Leave your setup for at least a week: the longer, the better! As the liquid drips off the string, a stalactite will form.



A stalactite forms slowly where the liquid drips off the string.

HOW IT WORKS

When the Epsom salt dissolves in the water it breaks down into particles called ions. These mix evenly and invisibly with the water molecules to make a strong solution. The string soaks up the solution all the way along its length through thousands of tiny hollow fibers. When the liquid falls off drop by drop, some of the ions in the solution join up, forming a solid crystal, while the water drips away. As more ions join together, the crystal grows.

